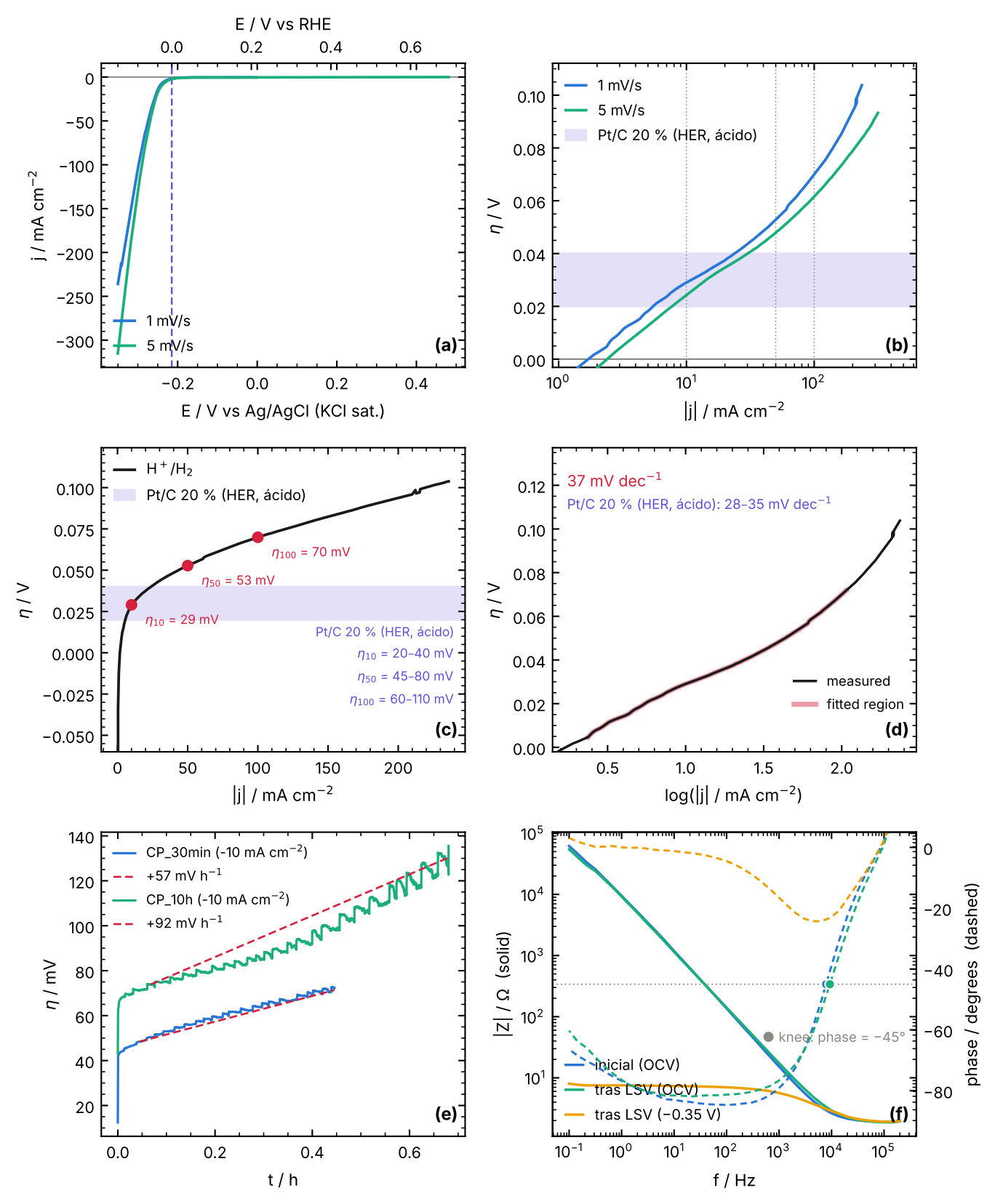

E / V vs RHE
0.0
0.2
0.4
0.6
0
−50
−100
−150
−200
−250
−300
j / mA cm^{-2}
1 mV/s
5 mV/s
(a)
−0.2
0.0
0.2
0.4
E / V vs Ag/AgCl (KCl sat.)
1 mV/s
5 mV/s
Pt/C 20 % (HER, ácido)
0.10
0.08
0.06
0.04
0.02
0.00
η / V
(b)
10^0
10^1
10^2
|j| / mA cm^{-2}
H$^+$/H$_2$
Pt/C 20 % (HER, ácido)
0.100
0.075
0.050
0.025
0.000
−0.025
−0.050
η / V
η_{100} = 70 mV
η_{50} = 53 mV
η_{10} = 29 mV
Pt/C 20 % (HER, ácido)
η_{10} = 20–40 mV
η_{50} = 45–80 mV
η_{100} = 60–110 mV
(c)
0
50
100
150
200
|j| / mA cm^{-2}
37 mV dec^{-1}
Pt/C 20 % (HER, ácido): 28–35 mV dec^{-1}
0.12
0.10
0.08
0.06
0.04
0.02
0.00
η / V
measured
fitted region
(d)
0.5
1.0
1.5
2.0
log(|j| / mA cm^{-2})
CP_30min (-10 mA cm^{-2})
+57 mV h^{-1}
CP_10h (-10 mA cm^{-2})
+92 mV h^{-1}
140
120
100
80
60
40
20
η / mV
(e)
0.0
0.2
0.4
0.6
t / h
10^5
10^4
10^3
10^2
10^1
|Z| / Ω (solid)
knee: phase = −45°
Inicial (OCV)
tras LSV (OCV)
tras LSV (−0.35 V)
(f)
0
−20
−40
−60
−80
phase / degrees (dashed)
10^{-1}
10^0
10^1
10^2
10^3
10^4
10^5
f / Hz